

. A transition element X has a configuration $\text{Ar}3d^4$ in its + 3 oxidation state. Its atomic number is

- (a) 25
- (b) 26
- (c) 22
- (d) 19

Answer: (a) 25

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

62. The transition element have a characteristic electronic configuration which can be represented as

- (a) $(n-2)s^2p^6d^{1-10}(n-1)s^2p^6ns^2$
- (b) $(n-2)s^2p^6d^{1-10}(n-1)s^1p^6d^1$ or d^1ns^1
- (c) $(n-1)s^1p^6d^{10}ns^2p^6nd^{1-10}$
- (d) $(n-1)s^2p^6d^{1-10}ns^1$ or ns^2

Answer: (d) $(n-1)s^2p^6d^{1-10}ns^1$ or ns^2

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

63. Number of unpaired electrons in Fe^{+++} (Z=26) is

- (a) 4
- (b) 5
- (c) 6
- (d) 3

Answer: (b) 5

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d^0 and d^{10} species are diamagnetic.

64. Of the ions Zn^{2+} , Ni^{2+} and Cr^{3+} [atomic number of $\text{Zn}=30$, $\text{Ni}=28$, $\text{Cr}=24$]

- (a) Only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured
- (b) All three are colourless
- (c) All three are coloured
- (d) Only Ni^{2+} is coloured and Zn^{2+} and Cr^{3+} are colourless

Answer: (a) Only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured

Explanation: The correct choice is Only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

65. Common oxidation state of scandium, a transition element is/are [atomic number of $\text{Sc} = 21$]

- (a) + 4
- (b) + 1
- (c) + 2 and + 3
- (d) + 4 and +1

Answer: (c) + 2 and + 3



Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

66. Which of the following is not correct about transition metals

- (a) Their melting and boiling points are high
- (b) Their compounds are generally coloured
- (c) They can form ionic or covalent compounds
- (d) They do not exhibit variable valency**

Answer: (d) They do not exhibit variable valency

Explanation: The correct choice is They do not exhibit variable valency. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

67. From +6 to +1 oxidation state is shown by the element of..... group

- (a) V-B
- (b) VI-B**
- (c) VII-B
- (d) VIII

Answer: (b) VI-B

Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

68. The electronic configuration of cobalt is

- (a) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^1, 4s^2$
- (b) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^7, 4s^2$**
- (c) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^3, 4s^2$
- (d) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^5, 4s^2$

Answer: (b) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^7, 4s^2$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before $(n-1)d$ electrons.

69. Out of all the known elements the number of transitional elements is

- (a) 80
- (b) 61**
- (c) 43
- (d) 38

Answer: (b) 61

Explanation: The correct choice is 61. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

70. Cigarette or gas lighter is made up of

- (a) Misch metal**
- (b) Alkali metal
- (c) Noble metal
- (d) None

Answer: (a) Misch metal

Explanation: The correct choice is Misch metal. This follows from the standard



periodic and electronic properties of d- and f-block elements tested in this question.

71. Bullet-proof steel alloy is prepared by using

- (a) Sc
- (b) Ni
- (c) Zr
- (d) Zn

Answer: (c) Zr

Explanation: The correct choice is Zr. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

72. In making gun-berrel, the steel used is

- (a) Fe-Mn
- (b) Fe-Cr
- (c) Fe-W
- (d) Ni-Mo

Answer: (d) Ni-Mo

Explanation: The correct choice is Ni-Mo. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

73. Neobium and tantalum metals are used in making surgical instruments because they are

- (a) Non-corrosive
- (b) Hard

- (c) Soft
- (d) All

Answer: (a) Non-corrosive

Explanation: The correct choice is Non-corrosive. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

74. To support tungstun filament in electric bulb, the steel used is

- (a) Cr
- (b) Ni
- (c) Mn
- (d) Mo

Answer: (d) Mo

Explanation: The correct choice is Mo. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

75. The elements belonging to the first transition series have atomic number from

- (a) 19 to 37
- (b) 22 to 32
- (c) 24 to 30
- (d) 21 to 30

Answer: (d) 21 to 30

Explanation: The correct choice is 21 to 30. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.



76. Which of the following elements does not belong to the first transition series

- (a) Fe
- (b) V
- (c) Ag**
- (d) Cu

Answer: (c) Ag

Explanation: The correct choice is Ag. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

77. Fe²⁺ shows

- (a) Ferromagnetism
- (b) Paramagnetism**
- (c) Diamagnetism
- (d) None of these

Answer: (b) Paramagnetism

Explanation: The correct choice is Paramagnetism. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

78. Zinc and mercury do not show variable valency like d-block elements because

- (a) They are soft
- (b) Their d- shells are complete**
- (c) They have only two electrons in the outermost subshell
- (d) Their d-shells are incomplete

Answer: (b) Their d- shells are complete

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

79. Cuprous ion is colourless while cupric ion is coloured because

- (a) Both have half filled p and d-orbitals
- (b) Cuprous ion has incomplete d-orbital and cupric ion has a complete d-orbital
- (c) Both have unpaired electrons in the d-orbitals
- (d) Cuprous ion has a complete d-orbital and cupric ion has an incomplete d-orbital**

Answer: (d) Cuprous ion has a complete d-orbital and cupric ion has an incomplete d-orbital

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d⁰ or d¹⁰ ions are generally colourless unless charge-transfer effects operate.

80. Transition metals are related to which block

- (a) s-block
- (b) p-block
- (c) d-block**
- (d) None of these

Answer: (c) d-block





Explanation: The correct choice is d-block. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

